7.4 Properties of Special Parallelograms

Name _____ Period ____ Date _____

Geometry 7.4 (30-minute plan) · Student sheet

I can…

- tell whether a statement about rhombuses, rectangles, and squares is always, sometimes, or never true.
- use the diagonals of rhombuses and rectangles to find measures.
- decide whether a quadrilateral in the coordinate plane is a rectangle, rhombus, or square.

Vocabulary · fill in the missing word

Rhombus — a _____ with four congruent sides

Rectangle — a _____ with four right angles

Square — a _____ with four congruent sides and four right angles

Part 1 · Notes and examples (fill in each note, then work the example)

Rhombus: A **rhombus** is a parallelogram with four _____.

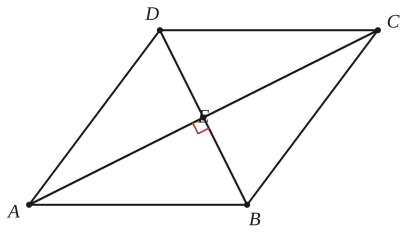
Rectangle: A **rectangle** is a parallelogram with four _____.

Square: A **square** is a parallelogram with four _____ and four _____.

Rhombus Diagonals and Opposite Angles Theorems: The diagonals of a rhombus are _____, and each diagonal _____ a pair of opposite angles.

Example 1

ABCD is a rhombus with diagonals meeting at E, and m∠ABC = 70°. Find m∠ABD and m∠AEB.



Rectangle Diagonals Theorem: The diagonals of a rectangle are _____.

Example 2

In rectangle ABCD, AC = 4x − 3 and BD = 2x + 9. Find x and AC.

Part 2 · Partner practice

Partner practice

In rectangle JKLM, JL = 5y − 8 and KM = 3y + 6. Find y and JL.